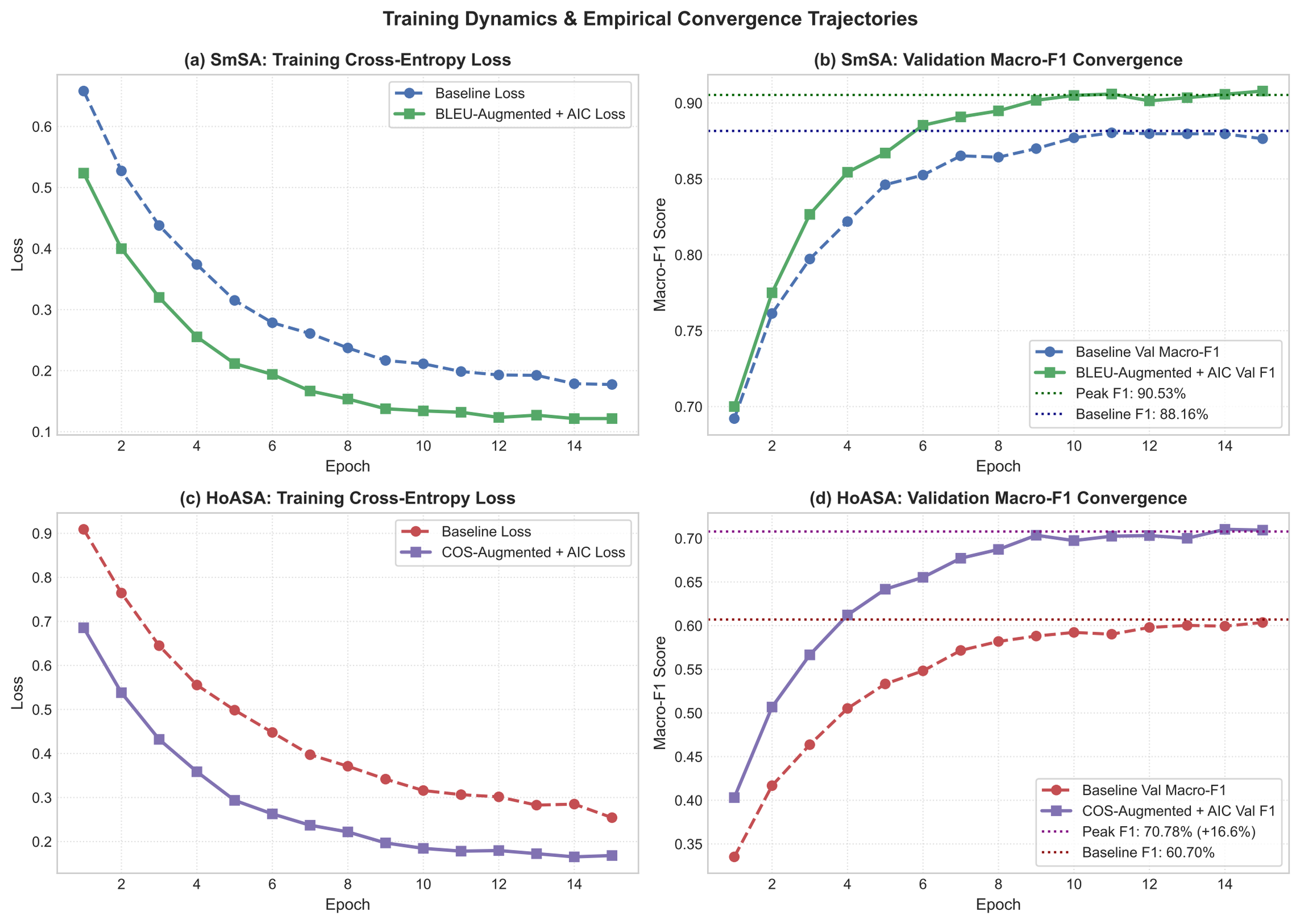
Training Dynamics & Empirical Convergence Trajectories
(a) SmSA: Training Cross-Entropy Loss
Baseline Loss
BLEU-Augmented + AIC Loss
Loss
Epoch
(b) SmSA: Validation Macro-F1 Convergence
Baseline Val Macro-F1
BLEU-Augmented + AIC Val F1
Peak F1: 90.53%
Baseline F1: 88.16%
Macro-F1 Score
Epoch
(c) HoASA: Training Cross-Entropy Loss
Baseline Loss
COS-Augmented + AIC Loss
Loss
Epoch
(d) HoASA: Validation Macro-F1 Convergence
Baseline Val Macro-F1
COS-Augmented + AIC Val F1
Peak F1: 70.78% (+16.6%)
Baseline F1: 60.70%
Macro-F1 Score
Epoch